

1. Administrative & Study Identification

Full Study Title: A Multicenter, Randomized, Open-label, Phase 2 Study Evaluating the Safety and Efficacy of BMS-986504 Monotherapy in Participants with Advanced or Metastatic Non-small Cell Lung Cancer (NSCLC) with Homozygous MTAP Deletion After Progression on Prior Therapies **Short Title/Acronym:** MountainTAP-9 **Protocol Number:** CA240-0009 **NCT Number:** NCT06855771 **Sponsor Name:** Bristol Myers Squibb **Investigational Product:** BMS-986504 (also known as MRTX1719) **Phase of Development:** Phase 2 **Medical Monitor:** Head of Global Submission Management - Clinical Trial (Bristol Myers Squibb)

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety and efficacy of BMS-986504 monotherapy in participants with advanced or metastatic Non-small Cell Lung Cancer (NSCLC) with homozygous MTAP deletion after progression on prior therapies. **Secondary Objectives:** To determine which of the two assigned doses of BMS-986504 is safer and/or more effective, and to evaluate overall survival and duration of benefit.

2.2 Background & Rationale Homozygous MTAP deletion occurs in approximately 13% of patients with NSCLC and is associated with shorter survival and acquired resistance to immune-targeted therapies. When MTAP is deleted, the metabolite MTA accumulates and binds to the enzyme PRMT5, which regulates cell survival. BMS-986504 is designed to selectively target and inhibit PRMT5 when it is bound to MTA (the PRMT5-MTA complex), leading to cell death in MTAP-deleted cancer cells while sparing normal tissue, thereby addressing the significant unmet need for this specific patient population.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Active Comparator (Two different dose levels of BMS-986504) **Blinding:** Open-label **Randomization:** Randomized

3.2 Planned Enrollment & Duration Target \$\$\$: Approximately 130 participants **Number of Sites:** Multicenter (Multiple locations worldwide, including UK, Australia, etc.) **Estimated Study Start:** 2024-2025 **Estimated Primary Completion:** Ongoing/Follow-up up to 3 years

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Histologically confirmed diagnosis of NSCLC and homozygous MTAP deletion detected in tumor tissue, and willingness to provide archival/fresh samples at screening for central MTAP status confirmation. 2. Advanced or metastatic NSCLC not amenable to curative therapies after progression on prior therapies at the time of enrollment (based on the American Joint Committee on Cancer, Ninth Edition). 3. At least 1 measurable lesion as per RECIST v1.1. 4. Documented radiographic disease progression on or after the most recent line of treatment. 5. Eastern Cooperative Oncology Group (ECOG) performance status of 0-1. 6. Participant must be ≥ 18 years of age. 7. Capability to swallow tablets intact (without chewing or crushing).

4.2 Exclusion Criteria 1. Active brain metastases or carcinomatous meningitis. 2. History of gastrointestinal disease or other conditions (e.g., uncontrolled nausea, vomiting, malabsorption syndrome) likely to alter absorption of study treatment or result in inability to swallow oral medications. 3. Prior treatment with a PRMT5 or MAT2A inhibitor. 4. Known severe hypersensitivity to study treatment and/or any of its excipients.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: BMS-986504 Dose 1 or Dose 2 (assigned via randomization) taken on specified days within 21-day cycles. **Route of Administration:** Oral (Tablets) **Duration of Treatment:** 21-day cycles administered continuously as long as the participant is benefiting from treatment and does not experience unacceptable side effects (until intolerable toxicity, death, or withdrawal of consent).

5.2 Concomitant & Prohibited Therapy Permitted: Standard oncology supportive care regimens. **Prohibited:** Other PRMT5 or MAT2A inhibitors, and treatments conflicting with the study's safety profile.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	Up to 28 days Informed Consent, Medical History, tumor tissue collection for central MTAP status confirmation
Treatment (Cycles)	Every 21 Days	Physical exam, Drug administration, Efficacy/Safety assessment	Safety Follow-Up
28 days post-dose	Safety evaluation, AE/SAE monitoring post-discontinuation	Survival Follow-Up	Every 3 months Survival tracking, subsequent anti-cancer therapies tracking (for up to 3 years)

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Efficacy and Safety of BMS-986504 monotherapy, assessed via Objective Response Rate (ORR) and incidence of treatment-emergent adverse events.

Measurement Method: Radiographic assessment per RECIST v1.1 and standard clinical safety evaluations.

7.2 Secondary & Exploratory Endpoints * Duration of Response (DOR) * Progression-Free Survival (PFS) * Overall Survival (OS)

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine collection of adverse events continuously throughout the treatment period and for 28 days following the last dose. **Serious Adverse Events (SAEs):** Expected standard 24-hour reporting timeline for oncology trials.

Data Monitoring Committee (DMC): Internal/External safety monitoring boards assessing dose-limiting toxicities to establish standard safety and efficacy boundaries.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy; Safety Set for all participants receiving at least one dose of BMS-986504. **Sample Size Justification:** Approximately 130 participants globally, sufficient to evaluate safety profiles and statistical significance of ORR between the two dose cohorts. **Handling of Missing Data:** Standard oncological methods applying censoring for progression and survival estimates.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Standard risk-based monitoring (onsite and remote) to ensure trial integrity and participant safety. **Audit Trail:** Mandatory eCRF entries, central laboratory confirmation of MTAP deletion status, and comprehensive data clarification processes.

11. Study Identifiers & Governance

Primary ID: CA240-0009 **Registry Number:** NCT06855771 **Sponsor/Lead Organization:** Bristol Myers Squibb **Study Title:** MountainTAP-9 **Official Regulatory Title:** A Multicenter, Randomized, Open-label, Phase 2 Study Evaluating the Safety and Efficacy of BMS-986504

Monotherapy in Participants with Advanced or Metastatic Non-small Cell Lung Cancer (NSCLC) with Homozygous MTAP Deletion After Progression on Prior Therapies

12. Clinical Framework (NSCLC Specific)

Primary Condition: Advanced or Metastatic Non-small Cell Lung Cancer (NSCLC) **Diagnosis Threshold:** Homozygous MTAP deletion **Medical Methodology:** Targeted PRMT5 inhibition (PRMT5-MTA complex) **Optimization Target:** Optimal dosing for efficacy and tolerability of BMS-986504

13. Study Procedures & Methodology

Management Pathway: Targeted Oncology Therapy Clinical Pathway **Education Protocol (HCP):** Site initiation visits and regular protocol safety training **Education Protocol (Patient):** Informed consent review, study medication oral administration guidelines **Quality Audit Mechanism:** Central confirmation of MTAP deletion via archival/fresh samples; RECIST v1.1 review

14. Observation & Follow-Up Schedule

Total Duration: Up to 3 years survival follow-up **Onsite Visit Cadence:** Every 21 days corresponding with treatment cycles **Remote Monitoring Frequency:** Regular data quality checks according to the monitoring plan **Checklist Review Interval:** Baseline and end of each 21-day cycle

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				